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# Simple Low-Cost White Noise Generator

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White noise is widely used during the testing of preamplifiers, filters, power amplifiers, etc. So white noise generators are required at least for the audio signal range. These generators exist in the market but usually are quite expensive. Here is a simple and low-cost white noise generator made with a Zener diode and three popular high-gain transistors that can be made easily.

This noise generator can be used individually for a wide variety of testing purposes. Also, it can be used with additional filters to produce such 'coloured' noises as pink noise, grey noise, blue noise, etc. The targeted applications for this noise generator are up to around 100kHz.

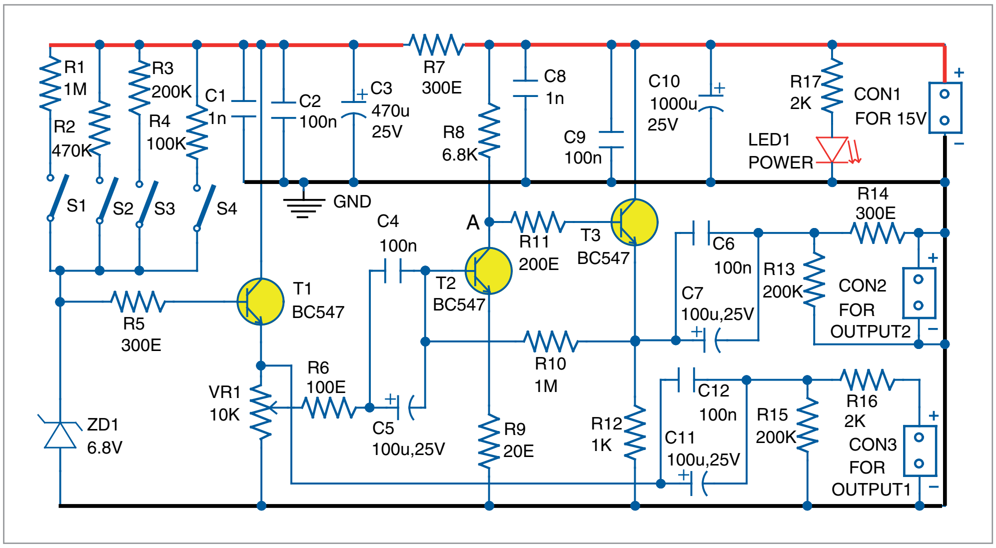


*Fig. 1: Circuit diagram of the white noise generator*

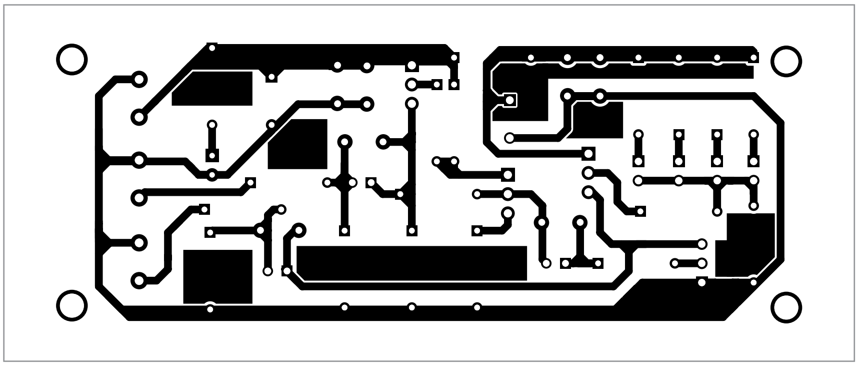

*Fig. 2: Actual-size PCB layout for the white noise generator*

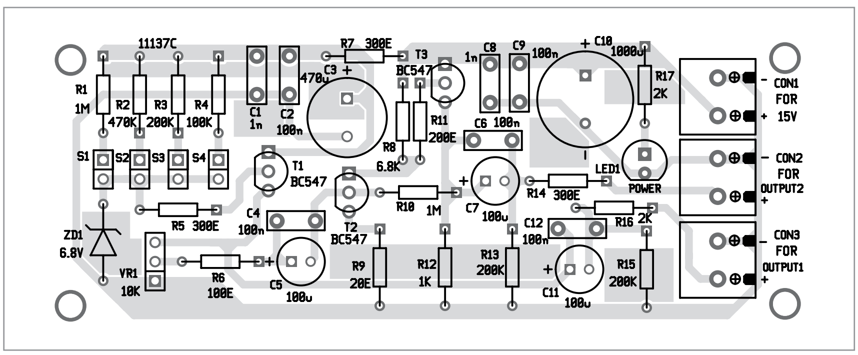

*Fig. 3: Components layout of the PCB*

## Circuit and working

Circuit diagram of the white noise generator is shown in Fig. 1. It is built around three BC547 transistors (T1 through T3) and a 6.8V Zener diode (ZD1). It requires a 15V DC power supply.

Breakdown voltage of Zener diode ZD1 should be lower than the power supply voltage Vcc, which is 15V in this case. Usually, there is no need to select the Zener diode according to the noise to be produced. But, if required, you can change the Zener diode, transistors, and the LED to suit the kind of noise to be produced.

The noise produced by the Zener diode depends on the current through the diode. Here we have four options for that current, which can be selected through switches S1,